

Mojtaba Keivanlou

Senior Hydraulic & Water Resources Engineer · Team Leader

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PROFESSIONAL SUMMARY

Senior Hydraulic and Water Resources Engineer with 16 years of experience delivering water engineering projects in Iran and Oman, alongside remote international work including US-regulated dam safety assessment. Currently Team Leader on one of Oman's largest ongoing flood protection programmes — the Batinah Flood Protection Master Plan for the Ministry of Agriculture, Fisheries & Water Resources, covering the North and South Al Batinah Governorates (15,000 km²) and encompassing several dams, governorate-wide flood protection channels, and stormwater systems.

Specialised in flood modelling and protection, climate adaptation, dam and spillway design, stormwater and sewage master planning, irrigation, water supply, and transient analysis — with a project record spanning national-scale flood risk programmes and multi-province flood hazard mapping. Distinguished by automation of engineering workflows, geospatial data pipelines, and AI-augmented project delivery alongside deep field expertise.

16+	50+	2
YEARS EXPERIENCE	PROJECTS DELIVERED	COUNTRIES + US REMOTE

AREAS OF EXPERTISE

🌊 Flood Modelling & Protection

1D/2D hydraulic modelling, flood hazard & risk mapping, masterplanning of protection schemes at national scale.

🏰 Dam Engineering

Spillway hydraulic design, dam breach analysis, dam safety assessment under FEMA guidelines, and site selection studies.

🌐 GIS & Spatial Analysis

Advanced QGIS and ArcGIS workflows, geospatial data pipelines, flood hazard mapping, and drone-based photogrammetry.

🛤 Road & Drainage Design

Hydraulic design of road drainage structures, culverts, bridges, and channels along major infrastructure corridors.

🔗 Stormwater & Sewage Networks

Gravity network design, urban drainage and sewage masterplanning, and network modelling using SewerGEMS and InfoWorks ICM.

📏 Surveying & Mapping

Total station (Leica), GNSS, UAV drone mapping, and photogrammetry for engineering survey and data collection.

☁ Hydrology & Climate Analysis

Rainfall-runoff modelling, frequency analysis, and hydrological studies incorporating climate change impact assessment.

💧 Water Supply & Irrigation

Urban and rural water supply networks, pressurised irrigation systems, and transmission pipelines with transient analysis.

📖 Engineering Automation

Python-based automation of hydraulic workflows, geospatial pipelines, and AI-augmented delivery.

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Renardet S.A. & Partners

Senior Hydraulic Engineer · Team Leader

Muscat, Sultanate of Oman

2022 – Present

Leading the hydraulic team on one of Oman's largest ongoing flood protection programmes. Currently Team Leader on the Batinah Flood Protection Master Plan (15,000 km², OMR 6M / ~USD 15.6M) for the Ministry of Agriculture, Fisheries & Water Resources, with parallel project delivery across Dhofar Governorate and major road infrastructure corridors.

Batinah Flood Protection Scheme — North & South Al Batinah Governorates
Feb 2023 – Present · Team Leader · OMR 6M (~USD 15.6M) · 15,000 km² study area
 - Leading all technical aspects: hydrology studies, hydraulic modelling, flood risk mapping
 - Dam site selection; design of dam hydraulic structures and spillway modelling (Flow 3D)
 - Integrated HEC-HMS / HEC-RAS / Python model; SewerGEMS & InfoDrainage for channel design
 - Client and stakeholder management, report writing and reviewing
 - Flood mapping under existing and proposed conditions for insurance and cost-benefit analysis

Majlas Dam Design Review — Quriyat
Jan 2026 – Present · Hydrology & Hydraulic Engineer
 - Review and update of hydrology report; revised rainfall and design flow values
 - Flood protection channel design downstream of dam in Quriyat
 - Techno-Economic and Flood-Hazard Assessment of the Wadi Majlas Scheme

Detailed Design of Flood Protection Schemes — Dhofar Governorate
Aug 2022 – Present · Hydraulic Engineer · Wilayat Salalah, Taqah, Mirbat, Rakhyoot & Dhalkoot
 - Hydrology studies and hydraulic modelling
 - Dam site selection and dam break analysis
 - Hydraulic strategy and meeting coordination

Upgrading of Existing Batinah Dual Carriageway
Jan 2022 – Mar 2023 · Hydraulic Engineer · 230 km road corridor
 - Comprehensive hydrology analysis and design of road drainage structures

Yankit Mixed Use Development
Sep 2022 – Present · Hydraulic Engineer
 - Design of road drainage, stormwater system and sewage network
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DuBois & King

Senior Hydraulic Engineer — Freelance Specialist

Vermont, USA (Remote)

2021

Dam safety assessment for Shadow Lake Dam under US federal and Vermont ANR standards. Comprehensive hydrology, dam breach modelling, and hazard classification per FEMA guidelines.

Shadow Lake Dam Safety Assessment — Glover, VT
Jan 2021 – Nov 2021
 - Comprehensive hydrology analysis and hydraulic design of dam
 - Dam breach analysis and hazard classification per FEMA guidelines
 - Delivered under US federal and Vermont ANR regulatory requirements

○ Sazeh Pardazi Iran

2020 – 2021

Water Resources Engineer — Drainage & Water Supply

Tehran, Iran

Flood hazard and risk mapping across three major Iranian provinces (409 km of river systems), river training and flood protection design, and water infrastructure projects including transmission pipelines and stormwater masterplanning.

Flood Hazard & Risk Mapping — Hormozgan Province

Jan 2020 – Sep 2020 · 127 km of river system

- Hydraulic modelling of the river system for flood hazard and risk mapping

River Training & Rehabilitation — Lorestan Province

Aug 2020 – Jan 2021 · 190 km of rivers

- Hydraulic modelling and detailed design of river training works

Flood Protection — Kerman Province

Jan 2021 – May 2021 · 92 km

- Hydraulic modelling and detailed design of flood protection structures

Water Transmission Pipeline — Karaj WWTP to Eshtehard Industrial Zone

Jun 2021 – Sep 2021

- Hydraulic design of the transmission pipeline
- Surge (water hammer) analysis

○ Hydrotech Toos (HTT)

2019 – 2020

Senior Hydraulic Engineer — Water Supply & Sewage

Mashhad, Iran

Senior engineer on water supply, sewage, and irrigation networks for Mashhad metropolitan area's developing zones. Included network optimisation and leakage detection across Khorasan Razavi Province.

Water Distribution Network Modelling — Khorasan Razavi Province

Jun 2019 – Jan 2020

- Hydraulic modelling of existing distribution network across multiple cities
- Leakage detection and network optimisation

Water Supply & Irrigation — Mashhad City

Jun 2019 – Jan 2020

- Hydraulic design of new water supply network and landscape irrigation pipeline revision

Sewage Network Modelling — Mashhad City

Jun 2019 – Jan 2020

- Hydraulic design of sewage network for developing areas

○ Earlier Positions — Iran

2010 – 2019

Project Manager · Hydraulic Engineer · GIS Technician

Multiple Provinces, Iran

Progressive nine-year career from GIS/CAD technician to Project Manager — spanning irrigation system design, provincial flood hazard mapping, and leading field survey teams with Leica instruments.

Majd Ab Shargh — Project Manager (Drainage)

Apr 2017 – Jun 2019

- Flood hazard and risk mapping — client interaction, team building, hydraulic modelling
- Field surveying with Leica total station instruments

Behin Ab Pazh — Hydraulic Engineer

Aug 2015 – Mar 2017

- Flood hazard and risk mapping — hydrologic analysis and hydraulic modelling

Zharfab Pay — Hydraulic Engineer

Oct 2013 – Jul 2015

- Design of pressurised irrigation systems

Abgostaran Sabz Dasht — GIS & CAD Draftsman

Oct 2010 – Dec 2013

- Comprehensive drainage project — east part of Mazandaran Province

EDUCATION

MSc — Hydraulic Structures Engineering

2013

Sari University of Agricultural Sciences and Natural Resources (SANRU), Iran

BSc — Water Sciences and Engineering

2009

University of Birjand, Iran

SKILLS & SOFTWARE

Hydraulic & Hydrological Modelling

HEC-RAS 1D-2D

HEC-HMS

SewerGEMS

InfoDrainage

HY-8

HEC-SSP

Storm & Sanitary Analysis

InfoWorks ICM

WaterGEMS

Hammer

Flow 3D

Delft3D

OpenFOAM

GIS & Spatial Analysis

QGIS (Advanced)

ArcGIS - ArcGIS Pro

Global Mapper

Google Earth Engine

PostGIS

QGIS Python Plugins

Programming & Automation

Python (Geospatial)

VBA

JAVA

R

PostgreSQL

MATLAB

Jython

CAD & Design

AutoCAD

Civil 3D

Subassembly Composer

Infraworks

Rhino

Revit

Survey & Field

Total Station (Leica)

GNSS - GPS

UAV - Drone Mapping

Photogrammetry

Project Management

MS Project

MBA

Data & Dev Tools

MS Office 365

Docker

GitHub

Power BI

AI Engineering Workflows

VS Code

Cursor

OpenAI Codex

Claude

LANGUAGES

Persian — Native

English — Fluent